CONTRACTIVE PROJECTIONS ON H^P FOR $0 < P < 1$

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Evidence is labelled Sketch: this proof has not been posted on arXiv or peer reviewed and should be checked by the reader. AI assistance was used in adapting and checking the argument and exposition.

> "It seems that the assumption $Pz^n \in H^\infty(\mathbb{T})$ in Theorem 3.8 arises from technical reasons. We conjecture that this assumption can be dropped. Specifically, we conjecture that if P is a contractive projection on $H^p(\mathbb{T})$ with $0 < p < 1$ satisfying $P1 = 1$, then it automatically maps bounded functions to bounded functions, and therefore $Pf = f(0)$ or P is the identity."

Source: Fu, Guo, and Li, "Contractive projections, conditional expectations, and idempotent coefficient multipliers on H^p spaces $(0 < p < 1)$," conjecture arXiv:2503.06615v2, after Theorem 3.8, pp. 15–16.

Answered: yes. Evidence: Sketch.

Conditional on: Fu, Guo, and Li, arXiv:2503.06615v2, Theorem 3.8.

Yes, conditional on their Theorem 3.8: every such projection is evaluation at zero or the identity.

Proof. Set

$$A_p(r) = \frac{1}{2\pi}\int_0^{2\pi} |1 + re^{i\theta}|^p \, \mathrm{d}\theta, \qquad r \geq 0.$$

Subharmonicity gives $A_p(r) \geq 1$. The binomial expansion for $0 \leq r < 1$ gives

$$A_p(r) = \sum_{k=0}^{\infty} \binom{p/2}{k}^2 r^{2k} = 1 + \frac{p^2}{4} r^2 + O_p(r^4).$$

Moreover, $0 \leq A_p(r) - 1 \leq C_p r^2$ for all $r \geq 0$: for $r \leq 1$ this follows from the nonnegative series (including its finite limit at $r = 1$), and for $r \geq 1$ from $A_p(r) \leq (1 + r)^p \leq 2^p r^p \leq 2^p r^2$.

Fix $f \in H^2(\mathbb{T})$ and write $g = Pf \in H^p(\mathbb{T})$. For every $\varepsilon > 0$ and $0 \leq \theta < 2\pi$, the hypotheses imply

$$\|1 + \varepsilon e^{i\theta} g\|_p^p \leq \|1 + \varepsilon e^{i\theta} f\|_p^p.$$

Averaging over θ, and evaluating g on any fixed circle of radius $0 < \rho < 1$, gives

$$\int_{\mathbb{T}} (A_p(\varepsilon |g(\rho\zeta)|) - 1) \, \mathrm{d}m(\zeta) \leq \int_{\mathbb{T}} (A_p(\varepsilon |f^*(\zeta)|) - 1) \, \mathrm{d}m(\zeta).$$

Indeed, the L^p integral on a fixed circle is at most the H^p norm, and the latter is the boundary integral for f. After division by ε^2, the left side converges by uniformity on the fixed circle, while dominated convergence applies on the right because $f^* \in L^2$ and $A_p(r) - 1 \leq C_p r^2$. The quadratic limit therefore yields

$$\int_{\mathbb{T}} |g(\rho\zeta)|^2 \, \mathrm{d}m(\zeta) \leq \|f\|_2^2.$$

Taking the supremum over ρ shows $g \in H^2$ and $P|_{H^2}$ is a contraction. It is still an idempotent fixing 1. A contractive idempotent on a Hilbert space is an orthogonal projection: if x belongs to its range and y to its kernel, then $\|x\|_2 \leq \|x + ty\|_2$ for every complex t, which forces $\langle x, y\rangle_{H^2} = 0$. Thus the restriction is an orthogonal projection on H^2.

For each fixed n, the coefficient functional $\lambda_n(f) = \widehat{Pf}(n)$ is continuous on H^p. Indeed, Taylor coefficients are continuous on H^p: the subharmonic point-evaluation bound on $|z| = 1/2$,

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followed by Cauchy's formula, gives $|\widehat{h}(n)| \leq C_{n,p}\|h\|_p$ for $h \in H^p$. The Duren–Romberg–Shields duality theorem Theorem 1 of Duren, Romberg, and Shields, *Linear functionals on H^p spaces with $0 < p < 1$*, doi:10.1515/crll.1969.238.32 (in the disc-algebra form stated in Lemma 3.2 of Fu, Guo, and Li) assigns to λ_n a unique $a_n \in A(\mathbb{D})$ with $\lambda_n(z^k) = \overline{\widehat{a_n}(k)}$ for every $k \geq 0$. By orthogonality, with the convention that the H^2 inner product is linear in the first argument,

$$\lambda_n(z^k) = \langle Pz^k, z^n\rangle_{H^2} = \langle z^k, Pz^n\rangle_{H^2} = \overline{\widehat{Pz^n}(k)}.$$

Thus a_n and Pz^n have identical Taylor coefficients and are the same analytic function; hence $Pz^n \in A(\mathbb{D})$.

The inclusion $Pz^n \in A(\mathbb{D}) \subset H^\infty$ verifies exactly the extra hypothesis of Theorem 3.8 of Fu, Guo, and Li. Their conclusion proves that P is evaluation at zero or the identity. □